

Reinforcement Learning for Portfolio Rebalancing

Review 2: Progress Report

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1. Introduction

An investor who holds several funds has to decide how often to bring the portfolio back to its planned mix. Rebalancing too often wastes money on brokerage, taxes and other charges, while rebalancing too rarely lets the portfolio drift into a riskier position. In practice most people follow a fixed calendar, for example every quarter, regardless of what the market is doing. Our project asks whether a reinforcement learning (RL) agent can learn *when* and *how much* to rebalance, and whether that leads to better risk-adjusted returns than fixed rules once real transaction costs are taken into account.

2. Objectives

- Build a realistic simulator of a weekly-rebalanced portfolio of Indian ETFs, including Indian trading costs.
- Compare an RL agent against standard rebalancing rules and classical portfolio optimisers.
- Evaluate everything strictly out of sample, on data the agent has never seen.

3. Methodology

Data. Weekly prices of seven NSE-listed ETFs (NIFTYBEES, JUNIORBEES, BANKBEES, PSUBNKBEES, INFRABEES, GOLDBEES and MON100) plus a cash position were downloaded from Yahoo Finance. The data is frozen as a snapshot so that every experiment uses identical inputs. It is split in time order into a training period (up to 2019), a validation period (2020 to 2021) and a test period (2022 onwards). The test period is kept locked until the final evaluation, and a short gap is left between periods to avoid information leaking across them.

Transaction costs. Every trade is charged STT, stamp duty, NSE and SEBI fees, GST and an allowance for slippage. The same cost model is applied to the agent and to every baseline.

Agent design. We use a hybrid approach. A classical optimiser (risk parity) proposes a target portfolio, and the RL agent only decides what fraction of the gap to that target to close each week, from 0 (do nothing) to 1 (fully rebalance). This keeps the action simple and easy to interpret. The reward each week is the log return after costs minus a penalty on losses (downside penalty, $\lambda = 5$), so the agent is discouraged from taking large risks. Training uses the PPO algorithm from Stable-Baselines3.

4. Work Completed

- Data pipeline with automatic download, cleaning and frozen snapshots.
- Indian transaction cost model and a portfolio simulator.
- Eleven baseline strategies, including buy-and-hold, equal-weight (monthly, quarterly and 5% band), minimum-variance, risk parity, HRP, mean-variance and the market index.
- Five sanity tests on the simulator. All five pass (details below).
- RL training pipeline, together with a web interface for running and viewing experiments.

Baseline results on the validation period (2020 to 2021), net of costs:

Strategy	Sharpe	CAGR	Max drawdown
Risk parity, monthly	0.73	19.1%	19.1%
Market index (NIFTYBEES)	0.64	20.8%	33.5%
Equal-weight, quarterly (benchmark)	0.63	19.2%	26.1%
Minimum-variance, monthly	0.62	16.3%	17.2%
Buy and hold	0.58	17.4%	26.1%
Mean-variance, monthly	0.29	10.9%	22.2%

Quarterly equal-weight rebalancing is the most common rule used in practice, so we treat its Sharpe ratio of 0.63 as the benchmark the agent has to beat. The market index gave a similar return but fell by about a third during the COVID crash, which shows why risk has to be considered alongside return.

Sanity tests. Before trusting any learning results we checked the simulator. Buy-and-hold values match a hand calculation, trading costs are charged exactly as expected, and an equal-weight action reproduces the equal-weight portfolio. We also trained the agent on shuffled returns, where there is nothing to learn, and it found no edge; when it was deliberately given next week's returns it performed far better. Together these confirm that the environment is correct and that the agent only learns when a real signal exists.

5. Preliminary Observations

A first trial run of the agent (PPO, two random seeds, 20,000 training steps each) completed successfully. On the validation period it followed the benchmarks closely and had a smaller drawdown during the 2020 crash. These results come from a small, untuned run and are not yet conclusive; they mainly confirm that the full pipeline works end to end.

6. Remaining Work

- Full training with more seeds and longer runs to obtain reliable estimates.
- Comparison of RL algorithms (PPO, A2C, SAC, TD3, DQN) and network designs.
- Comparison of reward settings (downside, variance and drawdown penalties) and risk aversion levels.
- Robustness checks under higher transaction costs and in different market regimes, and walk-forward testing.
- One-time evaluation on the locked test period, followed by the final report.